freely between the end of the valve stem and the rocker arm and the .010″ feeler gage will not pass through. Readjust the push rod, if necessary.

2. After the exhaust valve clearance has been adjusted, check the fuel injector timing (Section 14.2).

ENGINES WITH FOUR VALVE CYLINDER HEADS

The exhaust valve bridges must be adjusted and the adjustment screws locked securely at the time the cylinder head is installed on the engine. The necessary adjustment procedure is outlined in Section 1.2.2.

The exhaust valve bridge balance should be checked when a general valve adjustment is performed. After the bridges are balanced, adjust the valve clearance at the *push rod only. Do not disturb the exhaust valve bridge adjusting screw.*

All of the exhaust valves may be adjusted in firing order sequence during one full revolution of the crankshaft. Refer to the *General Specifications* at the front of the manual for the engine firing order.

Valve Clearance Adjustment (Cold Engine)

1. Remove the loose dirt from the valve rocker cover and remove the cover.

2. Place the governor speed control lever in the *idle* speed position. If a stop lever is provided, secure it in the *stop* position.

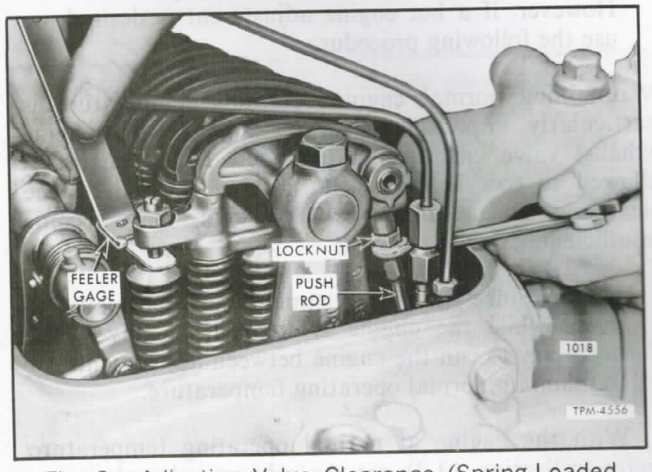


Fig. 2 - Adjusting Valve Clearance (Spring-Loaded Valve Bridge)

Check Exhaust Valve Clearance Adjustment

1. With the engine at 100° F (38° C) or less, check the valve clearance.

2. If a .012″ feeler gage (±.004″) will pass between the valve stem and the rocker arm bridge, the valve clearance is satisfactory. If necessary, adjust the push rod.

3. Rotate the crankshaft, manually or with the starting motor, until the injector follower is fully depressed on the particular cylinder to be adjusted.

> **NOTE:** If a wrench is used on the crankshaft bolt at the front of the engine, do not turn the crankshaft in a left-hand direction of rotation because the bolt could be loosen.

4. Loosen the exhaust valve rocker arm push rod locknut.

5. Place a .016″ feeler gage (J 9708-01) between the end of the exhaust valve stem and the valve bridge adjustment screw (spring-loaded bridge only) or between the valve bridge and the valve rocker arm pallet (unloaded bridge only) -- refer to Figs. 2 and 3. Adjust the push rod to obtain a smooth pull on the feeler gage.

6. Remove the feeler gage. Hold the push rod with a 5/16″ wrench and tighten the locknut with a 1/2″ wrench.

7. Recheck the clearance. At this time, if the adjustment is correct, the .015″ feeler gage will pass freely between the valve stem and the adjustment screw (spring-loaded bridge) or between the valve bridge and

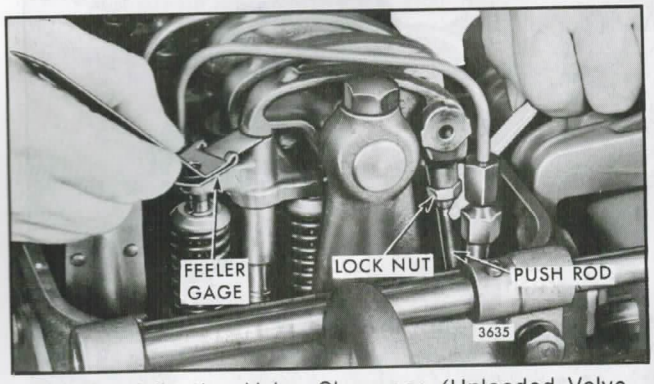


Fig. 3 - Adjusting Valve Clearance (Unloaded Valve Bridge)

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